



PROJECT REPORT

Design and Development of a Servo-Based Humanoid Robotic System

✦ Chapter 1: Introduction

1.1 Overview

Robotics is an advanced field of engineering that combines **mechanical design, electronics, and programming** to create intelligent machines. A **humanoid robot** is designed to resemble the human body structure and perform movements similar to humans.

This project focuses on designing and building a **servo motor-based humanoid robot** capable of performing controlled movements using multiple joints.

1.2 Purpose of the System

The main purpose of this system is:

- To simulate **human-like movement**
 - To understand **robotic joint control systems**
 - To develop a **low-cost humanoid prototype**
 - To apply knowledge of **mechanical and embedded systems**
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1.3 Objectives

- Design a **multi-joint robotic structure**
 - Implement **servo motor control system**
 - Achieve **stable standing posture**
 - Enable **arm and leg movement**
 - Prepare system for future **AI and sensor integration**
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1.4 Scope of the Project

- Manual + programmable robotic control
 - Indoor experimental robot
 - Educational and research usage
 - Future upgrade: walking, obstacle detection, AI
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Chapter 2: Methodology

2.1 Development Approach

The project follows a **hardware-based prototyping methodology**:

1. Design robot structure
 2. Select components
 3. Assemble mechanical parts
 4. Connect electronics
 5. Test movement
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2.2 System Design Process

- Sketch humanoid design
 - Divide into parts:
 - Head
 - Arms
 - Legs
 - Torso
 - Assign servo motors to joints
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2.3 Working Principle

The robot operates using **servo motors**, which rotate to specific angles based on control signals.

- Each joint = 1 servo motor

- Controller sends PWM signals
 - Servo moves to defined angle
 - Combined movement = humanoid motion
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Chapter 3: System Analysis

3.1 Degrees of Freedom (DOF)

Your robot includes:

- Shoulder → 2 DOF
- Elbow → 1 DOF
- Hip → 2 DOF
- Knee → 1 DOF
- Ankle → 1 DOF

 Total: **10–14 DOF (approx.)**

3.2 Mechanical Analysis

- Frame made of **metal/acrylic plates**
 - Joints connected using screws and brackets
 - Base designed for **balance and stability**
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3.3 Electrical Analysis

- Servo motors require **5V–6V power**
 - Each motor consumes **high current**
 - External power supply recommended
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3.4 System Limitations

- No balance algorithm
- No sensors yet

- Manual wiring complexity
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Chapter 4: System Design

4.1 Mechanical Design

Based on your images:

- **Humanoid shape structure**
 - Arms and legs with joint connections
 - Wide feet for stability
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4.2 Block Diagram

Power Supply → Controller → Servo Motors → Robot Movement

4.3 Circuit Design

Components connected as:

- Red → VCC (5V)
 - Brown/Black → GND
 - Yellow/Orange → Signal
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Chapter 5: Implementation

5.1 Hardware Components (IMPORTANT 🔥)

1. Servo Motors

- Type: MG996R / SG90 (based on your design)
 - Function: Joint movement
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2. Microcontroller (Recommended)

- Arduino UNO / NodeMCU

- Controls servo signals
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3. Power Supply

- 5V–6V battery / adapter
 - Required for multiple servos
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4. Frame Material

- Metal sheets / Acrylic plates
 - Provides structure
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5. Jumper Wires

- Used for connections
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6. Servo Brackets & Screws

- Connect joints
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7. Breadboard / Driver (Optional)

- PCA9685 servo driver (recommended for many servos)
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5.2 Assembly Process

1. Build leg structure
2. Attach servo motors
3. Build arm system
4. Connect torso
5. Install head
6. Wiring connections

5.3 Software (Optional)

Example Arduino logic:

- Define servo pins
 - Set angles
 - Move joints
-

Chapter 6: Testing

6.1 Testing Types

✓ Movement Test

- Check each servo rotation

✓ Stability Test

- Robot standing balance

✓ Power Test

- Voltage stability
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6.2 Results

- Robot successfully moves joints
 - Stable in standing position
 - Needs improvement in walking
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Chapter 7: Conclusion

7.1 Summary

This project successfully demonstrates the design and construction of a **humanoid robotic system using servo motors**. The robot can perform basic movements and serves as a foundation for advanced robotics development.

7.2 Future Improvements

- Add **Arduino control system**
 - Implement **walking algorithm**
 - Add **ultrasonic sensors**
 - Add **camera + AI**
 - Wireless control (Bluetooth/WiFi)
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BONUS (For High Marks 100)

Add this section:

Applications

- Education & research
- Industrial automation
- Service robots
- AI robotics

2. CIRCUIT DIAGRAM (EXPLANATION)

Basic Connection

Each Servo has 3 wires:

Wire Color	Connection
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Red	5V (Power)
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Brown/Black	GND
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Yellow/Orange	Arduino Digital Pin
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Example Connection (4 Servos)

Servo	Arduino Pin
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Servo 1 (Right Arm)	D3
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Servo	Arduino Pin
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Servo 2 (Left Arm)	D5
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Servo 3 (Right Leg)	D6
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Servo 4 (Left Leg)	D9
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 IMPORTANT (VERY IMPORTANT 🔥)

 Do NOT power many servos from Arduino directly

 Use **external 5V battery or power module**

Correct Power Setup:

Battery (5V–6V)



Servo VCC (ALL)



Common GND → Arduino GND

 Recommended Upgrade

 Use **PCA9685 Servo Driver**

- Controls up to 16 servos
 - Stable power
 - Better performance
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 3. ARDUINO CODE (WORKING)

 Simple Servo Control Code

```
#include <Servo.h>
```

```
// Create servo objects
```

```
Servo rightArm;
```

```
Servo leftArm;
```

```
Servo rightLeg;
```

```
Servo leftLeg;

void setup() {
  // Attach servos to pins
  rightArm.attach(3);
  leftArm.attach(5);
  rightLeg.attach(6);
  leftLeg.attach(9);
}

void loop() {

  // Move arms up
  rightArm.write(90);
  leftArm.write(90);
  delay(1000);

  // Move arms down
  rightArm.write(0);
  leftArm.write(0);
  delay(1000);

  // Move legs forward
  rightLeg.write(45);
  leftLeg.write(45);
  delay(1000);

  // Reset legs
  rightLeg.write(0);
  leftLeg.write(0);
  delay(1000);
}
```

Advanced Idea (Next Level)

You can add:

- Bluetooth control (HC-05)
- Mobile app control

- Walking algorithm
- Sensor-based movement